

Josselin Renan

3D Software Engineer · real-time, simulation, XR

3+ years of R&D engineering at CEA List: building real-time 3D tools, physics simulation and industrial digital twins. Looking for a 3D software engineer position (simulation, real-time, XR).

Available from January 2027

LinkedIn

+33 6 95 74 40 30 josselin.renan@gmail.com Location: Paris area, open to anywhere in France

SKILLS

Engines & platforms
NVIDIA Omniverse (Kit), OpenUSD, Unity

XR
OpenXR, VR, AR (Vuforia)

Simulation
Physics engine integration, digital twins

Languages
Python, C++, C#, C

Architecture
OOP, ECS

English
TOEIC 905/990

EDUCATION

EFREI Paris

Engineering degree (Master's level), major in 3D and immersive technologies

2018 – 2023

INTERESTS

Film & animation

Running, rugby

PROFESSIONAL EXPERIENCE

Research engineer

11/2023 – 11/2026

CEA List · Interactive Simulation Laboratory

Hired after my final-year internship (04/2023 – 11/2023): exploring NVIDIA Omniverse with a view to integrating XDE.

XDE physics engine in NVIDIA Omniverse

- Development of Omniverse Kit extensions integrating the XDE physics engine (developed at CEA), from low-level implementation to the authoring interface for realistic physics scenes (robots, cables, joints...).
- Custom OpenUSD schemas describing physics and interactions.

Industrial digital twins

- Design of digital twins for critical environments: nuclear site decommissioning.
- VR assembly scene: assembling parts while interacting with a robot; machining scenes.

Demonstrations

- Demonstrator with Renault: real-time physical interaction with a vehicle in XR, shown at Laval Virtual and XR Days (2025).
- Demonstrator of the XDE platform at Laval Virtual (2026): robotics, drone inside the WEST tokamak, cables, sensors, assembly / disassembly, avatars.
- Robotics demonstrator in VR at Cap Digital (2026), as part of the 2MORDOR platform.

Omniverse Kit · OpenUSD · OpenXR · XDE · Python · C++

VR development internship

11/2021 – 04/2022

Virtual Rangers

- Immersive CAVE experience for CFL (Luxembourg railways): interactive exploration of 3D building models.
- Design of a playful virtual reality experience: broomstick flight through a fantasy world.